



DUNMAN SECONDARY SCHOOL

CANDIDATE
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CLASS

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NUMBER

PRELIMINARY EXAMINATION 2025 SECONDARY 4 EXPRESS / 5 NORMAL ACADEMIC

ADDITIONAL MATHEMATICS

Paper 2

4052/02

25 August 2025
2 hours 15 minutes

SOLUTIONS

Question	Answer
1a	$h = 7 + 20t - 5t^2$
	$= -5(t^2 - 4t) + 7$
	$= -5[(t - 2)^2 - 4] + 7$
	$= -5(t - 2)^2 + 27$
	maximum height = 27
	OR
	$\frac{dh}{dt} = -10t + 20$
	$-10t + 20 = 0$
	$t = 2$
	When $t = 2$, $h = 7 + 20(2) - 5(2)^2 = 27$ m
1b	$x^2 + 6x + k = k(x - 1)$
	$x^2 + (6 - k)x + 2k = 0$
	$(6 - k)^2 - 4(1)(2k) < 0$
	$36 - 12k + k^2 - 8k < 0$
	$k^2 - 20k + 32 < 0$
	$(k - 2)(k - 18) < 0$
	$\therefore 2 < k < 18$
2a	$\tan A = \frac{4}{3}$
	$\tan 2A = \frac{2\left(\frac{4}{3}\right)}{1 - \left(\frac{4}{3}\right)^2}$
	$= -\frac{24}{7}$

Question	Answer
2b	$\sin B = \frac{12}{13}, \cos B = \frac{5}{13}$
	$\sin\left(B + \frac{3\pi}{2}\right) = \sin B \cos \frac{3\pi}{2} + \sin \frac{3\pi}{2} \cos B$
	$= \frac{12}{13} \times 0 + (-1) \times \frac{5}{13}$
	$= -\frac{5}{13}$
	OR
	$\sin\left(B + \frac{3\pi}{2}\right) = \sin\left(2\pi - \frac{\pi}{2} + B\right)$
	$= \sin\left(-\frac{\pi}{2} + B\right) = -\sin\left(\frac{\pi}{2} - B\right)$
	$= -\cos B = -\frac{5}{13}$
2c	$\cos B = 2 \cos^2 \frac{B}{2} - 1$
	$\frac{5}{13} = 2 \cos^2 \frac{B}{2} - 1$
	$2 \cos^2 \frac{B}{2} = \frac{18}{13}$
	$\cos^2 \frac{B}{2} = \frac{9}{13}$
	$\cos \frac{B}{2} = \frac{3}{\sqrt{13}} \text{ or } -\frac{3}{\sqrt{13}} (\text{rej})$

Question	Answer
3a	$\angle PQC = \angle PSQ$ (Alternate segment theorem)
	$\angle PQC = \angle PBC$ (angles in same segment)
	Hence $\angle PBC = \angle PSQ$
	By alternate angles of parallel lines, BC is parallel to QS
3b	$\angle PQC = \angle PSQ$ (Alternate segment theorem)
	$\angle PMQ = \angle QMS$ (Common angle)
	$\triangle PQM$ is similar to $\triangle QSM$ (AA Similarity Test)
3c	Since $\triangle PQM$ is similar to $\triangle QSM$, $\frac{QM}{SM} = \frac{PM}{QM}$
	$(QM)^2 = PM \times SM$
4a	$\frac{d}{dx}(\cot x) = \frac{d}{dx}\left(\frac{\cos x}{\sin x}\right)$
	$= \frac{\sin x(-\sin x) - \cos x(\cos x)}{\sin^2 x}$
	$= \frac{-\sin^2 x - \cos^2 x}{\sin^2 x}$
	$= -\frac{1}{\sin^2 x}$
	$= -\operatorname{cosec}^2 x$ (Shown)
	OR

	$\frac{d}{dx}(\cot x) = \frac{d}{dx}(\tan x)^{-1}$
	$= -(\tan x)^{-2}(\sec^2 x)$
	$= -\left(\frac{\sin x}{\cos x}\right)^{-2}\left(\frac{1}{\cos^2 x}\right)$
	$= -\left(\frac{\cos^2 x}{\sin^2 x}\right)\left(\frac{1}{\cos^2 x}\right)$
	$= -\frac{1}{\sin^2 x}$
	$= -\operatorname{cosec}^2 x \text{ (Shown)}$
4b	$\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} -\cot^2 x \, dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} 1 - \operatorname{cosec}^2 x \, dx$
	$= \left[x + \cot x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$
	$= \left[\frac{\pi}{3} + \cot \frac{\pi}{3} \right] - \left[\frac{\pi}{4} + \cot \frac{\pi}{4} \right]$
	$= \frac{\pi}{3} + \frac{1}{\sqrt{3}} - \frac{\pi}{4} - 1$
	$= \frac{\pi}{12} + \frac{1}{\sqrt{3}} - 1 \text{ or } -0.161$

Question	Answer
5a	General term $= \binom{8}{r} x^{8-r} \left(-\frac{k}{x^3}\right)^r$
	$= \binom{8}{r} x^{8-r} (-1)^r k^r (x^{-3})^r$
	$= \binom{8}{r} (-1)^r k^r x^{8-4r}$
	When $8-4r=0$, $r=2$
	$\binom{8}{2} (-1)^2 k^2 x^{8-4(2)} = 7$
	$28k^2 = 7$
	$k^2 = \frac{1}{4}$
5b	$k = \frac{1}{2}$
	When $8-4r=-4$, $r=3$
	$T_4 = \binom{8}{3} (-1)^3 \left(\frac{1}{2}\right)^3 x^{-4}$
	$= -7x^{-4}$
	$\left(1+x^4\right)\left(x-\frac{k}{x^3}\right)^8 = \left(1+x^4\right)(\dots+7-7x^{-4}+\dots)$
	$= \dots+7-7+\dots=\dots+0+\dots$
	There is no constant term in the expansion. (Shown)

Question	Answer
6a	$AM = 2 \cos \theta$
	$BM = 2 \sin \theta$
	$BC = 4 \sin \theta$
	Total area = $(4 \sin \theta)^2 + \frac{1}{2}(2 \cos \theta)(4 \sin \theta)$
	$= 16 \sin^2 \theta + 4 \sin \theta \cos \theta$
6b	$16 \sin^2 \theta + 4 \sin \theta \cos \theta$
	$= 8(2 \sin^2 \theta) + 2(2 \sin \theta \cos \theta)$
	$= 8(1 - \cos 2\theta) + 2(\sin 2\theta)$
	$= 2 \sin 2\theta - 8 \cos 2\theta + 8$
6c	$R = \sqrt{2^2 + 8^2} = \sqrt{68}$
	$\tan \alpha = \frac{8}{2}$
	$\alpha = 76.0^\circ$
	$2 \sin 2\theta - 8 \cos 2\theta = \sqrt{68} \sin (2\theta - 76.0^\circ)$
7a	$LHS = \frac{\cos^2 x - \sin^2 x}{1 + 2 \sin x \cos x}$
	$= \frac{\cos^2 x - \sin^2 x}{\sin^2 x + \cos^2 x + 2 \sin x \cos x}$
	$= \frac{(\cos x + \sin x)(\cos x - \sin x)}{(\cos x + \sin x)^2}$
	$= \frac{\cos x - \sin x}{\cos x + \sin x}$
	$= \frac{1 - \tan x}{1 + \tan x} = RHS$

Question	Answer
7b	$\frac{1 - \tan x}{1 + \tan x} = \frac{2}{3} \tan x$
	$3(1 - \tan x) = 2 \tan x + \tan^2 x$
	$2 \tan^2 x + 5 \tan x - 3 = 0$
	$(2 \tan x - 1)(\tan x + 3) = 0$
	$\tan x = \frac{1}{2} \text{ or } -3$
	$x = 0.464 \text{ or } 3.61$
	$x = 1.89 \text{ or } 5.03$
8a	$ \begin{array}{r} x^2 - \frac{3x}{2} - 2 \\ 2x - 4 \overline{) 2x^3 - 7x^2 + 2x + 3} \\ \underline{2x^3 - 4x^2} \\ -3x^2 + 2x + 3 \\ \underline{-3x^2 + 6x} \\ 3 - 4x \\ \underline{-8 + 4x} \\ -5 \end{array} $

Question	Answer
8b	$2x^3 - 7x^2 + 2x + 3 = 0$
	Let $f(x) = 2x^3 - 7x^2 + 2x + 3$
	When $x = 3$, $f(3) = 2(3)^3 - 7(3)^2 + 2(3) + 3 = 0$
	By factor theorem, $(x - 3)$ is a factor of $f(x)$.
	$(x - 3)(2x^2 - x - 1) = 0$
	$(x - 3)(2x + 1)(x - 1) = 0$
	$x = -\frac{1}{2}$, 1 or 3
8c	$3y^6 + 2y^4 - 7y^2 + 2 = 0$
	$3 + \frac{2}{y^2} - \frac{7}{y^4} + \frac{2}{y^6} = 0$
	$3 + 2\left(\frac{1}{y^2}\right) - 7\left(\frac{1}{y^4}\right) + 2\left(\frac{1}{y^6}\right) = 0$
	$3 + 2\left(\frac{1}{y^2}\right) - 7\left(\frac{1}{y^2}\right)^2 + 2\left(\frac{1}{y^2}\right)^3 = 0$
	$\therefore x = \frac{1}{y^2}$, Kun Ye is correct

Question	Answer
9a	Height = $3x^2$
	Base = $2x$
	Area = $\frac{1}{2}(2x)(3x^2) = 3x^3$
9b	$\frac{dy}{dx} = 6x$
	$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$
	$3 = 6x \times \frac{dx}{dt}$
	$3 = 6x \times \frac{dx}{dt}$
	$\frac{dx}{dt} = \frac{1}{2x}$
	When $x = 4$, $\frac{dx}{dt} = \frac{1}{8}$
	Rate of change of $PQ = \frac{1}{4}$ unit/s
9c	$\frac{dA}{dx} = 9x^2$
	$\frac{dA}{dt} = \frac{dA}{dx} \times \frac{dx}{dt}$
	$\frac{dA}{dt} = 9x^2 \times \frac{1}{8}$
	When $x = 4$, $\frac{dA}{dt} = 18$ unit ² /s

Question	Answer
10a	$y\text{-coordinate of centre} = \frac{-1+9}{2} = 4$
10b	Let the x-coordinate of the centre be a ,
	$(x-a)^2 + (y-4)^2 = 25$
	From the line, $y = \frac{4x+9}{3}$
	$(x-a)^2 + \left(\frac{4x+9}{3} - 4\right)^2 = 25$
	$(x-a)^2 + \left(\frac{4}{3}x - 1\right)^2 = 25$
	$x^2 - 2ax + a^2 + \frac{16}{9}x^2 - \frac{8}{3}x + 1 = 25$
	$\frac{25}{9}x^2 + \left(-2a - \frac{8}{3}\right)x + a^2 - 24 = 0$
	Since the line is a tangent, $b^2 - 4ac = 0$
	$\left(-2a - \frac{8}{3}\right)^2 - 4\left(\frac{25}{9}\right)(a^2 - 24) = 0$
	$4a^2 + \frac{32}{3}a + \frac{64}{9} - \frac{100}{9}a^2 + \frac{2400}{9} = 0$
	$36a^2 + 96a + 64 - 100a^2 + 2400 = 0$
	$-64a^2 + 96a + 2464 = 0$
	$2a^2 - 3a - 77 = 0$
	$(2a+11)(a-7) = 0$
	$a = 7$ or -5.5 (rej)

Question	Answer
10c	Distance between point and centre $= \sqrt{(7-3)^2 + (4-0)^2}$
	$= 5.66$ unit
	Since the distance is longer than the radius, the point lies outside.
11a	$CX = \sqrt{(30-x)^2 + 20^2}$
	$= \sqrt{x^2 - 60x + 1300}$
	$T = \frac{x}{4} + \frac{1}{2}\sqrt{x^2 - 60x + 1300}$
11b	$\frac{dT}{dx} = \frac{1}{4} + \frac{1}{2} \left[\frac{1}{2} (x^2 - 60x + 1300)^{-\frac{1}{2}} (2x - 60) \right]$
	$= \frac{1}{4} + \frac{2x - 60}{4\sqrt{x^2 - 60x + 1300}}$
11c	$\frac{1}{4} + \frac{2x - 60}{4\sqrt{x^2 - 60x + 1300}} = 0$
	$\sqrt{x^2 - 60x + 1300} + 2x - 60 = 0$
	$\sqrt{x^2 - 60x + 1300} = 60 - 2x$
	$x^2 - 60x + 1300 = 3600 - 240x + 4x^2$
	$3x^2 - 180x + 2300 = 0$
	$x = \frac{180 \pm \sqrt{180^2 - 4(3)(2300)}}{2(3)}$
	$x = 41.5$ (rej) or 18.5